

LAB 4: ENTERPRISE WINDOWS SERVER ADMINISTRATION

CNIT24200-03

Group 3

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BUSINESS SCENARIO

WoodRock had implemented two Nutanix Acropolis Hypervisors (AHV), each had hosted separate clusters, where a multitude of Virtual Machines (VM) resided. The VMs offered increased scalability for employees to sell wood, as well as lowered infrastructure cost, which skyrocketed WoodRock's revenue for the quarter. Nutanix offered WoodRock the ability to create VMs, where several Windows 11 and Windows Server 2025 images were booted. G03SRV01VM, which was the parent-domain server in the forest, and the G03SRV03VM, which was a child-domain to the parent, servers held all the company's users for each department.

As the company expanded rapidly, WoodRock was in dire need for additional organization within company roles. WoodRock decided to create three separate Nutanix projects, which was WindowsServer2025VMConfigure, Windows11VMConfigure, and ThirdProject. Where company roles like Executive, Accounting, and Administrators were allocated, each had been limited to the respective authorization policies created. In order for WoodRock's employee users to have gained access to the projects, Nutanix Prism Central was first required to join the Active Directory domains through Lightweight Directory Access Protocol (LDAP) authentication, which included both group03.c24200.cit.lcl and c242-03-a.group03.c24200.cit.lcl domains.

Organization was not WoodRocks only issue, the company needed an answer for data share to authorized users, which was where True Network Attached Storage (TrueNAS) came into the picture. A TrueNAS image was booted on a G03TRUENAS VM in Nutanix Prism Central, which was pointed to the group03.c24200.cit.lcl Domain Naming Server (DNS) domain. TrueNAS gave WoodRock the opportunity to create two shared dataset folders with the domain

controller, GeneralSet03 and BackupSet03, both of which were stored in a newly created TrueNAS storage pool. BackupSet03 was a valuable asset as a backed-up company data storage. The company feared the possibility of mass data loss, WoodRock required server backups for all user and domain controller database files. Since G03TRUENAS was authenticated against G03SRV01VM, the server domain controller was able to schedule backups to the BackupSet03 folder because the company used Windows Backup Server tool and a script.

WoodRock sought true expansion for the future, which meant data had to be replicated across multiple physical locations, and administrative abilities needed to be separated. A new subnet range was created for the company on the G03NUTXSRV01 cluster, where two new VMs, G03SRV04 and G03S2WIN11, were hosted. The new subnet was created under 44.38.103.0/24, and joined to a newly created site. WoodRock was now able to replicate across multiple sites in the forest, which gave the company an opportunity to blossom out of location restrictions. Additionally, a server was created in the original subnet for WoodRock to separate administrative roles in the domain controllers. The Nutanix VM G03SRV05 was created on the G03NUTXSRV01 cluster to inherit the Infrastructure master, Relative Identifier (RID) master, and Domain Naming Master Flexible Single Master Operator (FSMO) roles from G03SRV01VM.

WoodRock had one final wish in company horizons expansion: made resources available over the web for different authorization levels of users. An additional VM was created on the G03NUTXSRV02 cluster called G03SRV06, where three web pages would be created with the Internet Information Services (IIS) server tool. One page for the general public was made to access company wood services, another page was configured for domain users only to have accessed company data and resources, finally a page for company employees located at the

original site only to have accessed important company information. The diagrams below represented the topology of the company before WoodRocks attempted expansion.

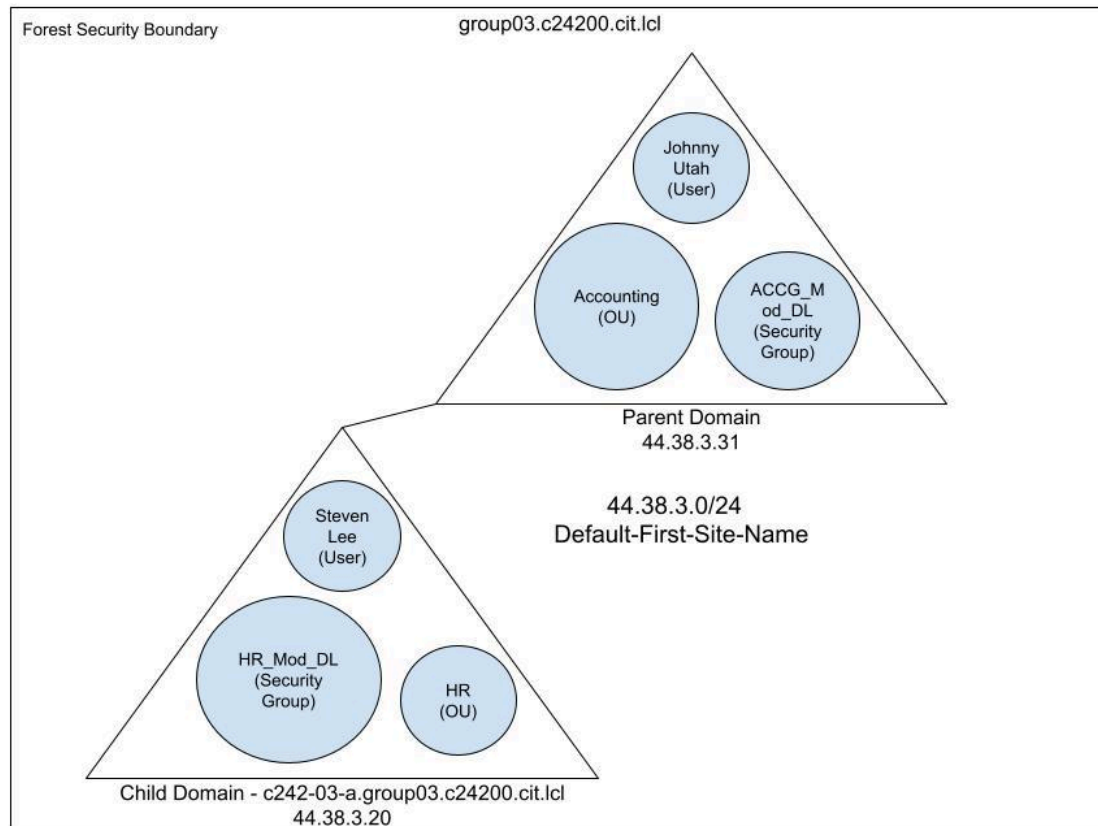


Figure 1.) Logical model of WoodRock's forest pre-execution of procedures

The logical model represented the parent domain and the child domain with the corresponding Organizational Units (OU), security groups, and users. Accounting was a major company priority for growth, hence, remained as an OU on the parent domain. Accounting OU allowed users such as Johnny Utah, who was referred to as an accounting prodigy, gained modification access to OU data through the ACCG_Mod_DL security group. Now Steven Lee, a member of the Human Resources (HR) department, was an exceptional employee for the child domain's HR OU. Steven Lee's talents gave him the opportunity to be part of the HR_Mod_DL modification

security group. WoodRock had implemented a singular site at the time, which was under the name Default-First-Site-Name and included only a singular subnet range.

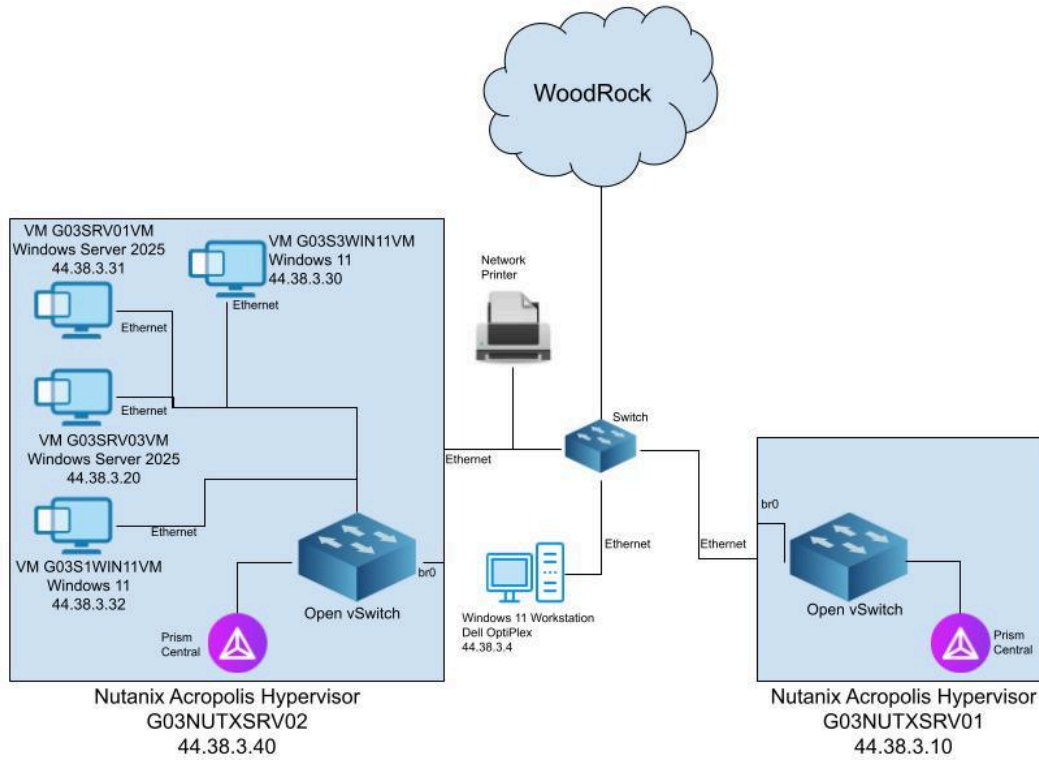


Figure 2.) Physical model of WoodRock topology pre-execution of procedures

WoodRock was dependent on infrastructure, which prompted the implementation of two Nutanix Acropolis Hypervisors alongside the respective clusters. The AHVs were booted on both Dell Pro Tower Plus machines. WoodRock was yet to have utilized the newly configured G03NUTXSRV01 cluster, though G03NUTXSRV02 hosted four VMs connected to a vSwitch via ethernet. Both hypervisors connected to a central switch via ethernet, which also connected to the Dell OptiPlex 5060 and the network printer. Finally, the switch connected the entire topology to the WoodRock, or the Computer Information Technology (CIT), network.

PROCEDURES

From a technical point of view, WoodRock took steps to expand the infrastructure. First, Nutanix Prism Central was added to the Active Directory environment and authenticated, which created a centralized management system. Next, TrueNAS storage appliance was applied, which provided secure Server Message Block (SMB) shares for general organizational data and specialized repositories for automated disaster recovery. As the business infrastructure grew, the network expanded with new subnets and sites to host additional redundant Domain Controllers, such as G03SRV04 and G03SRV05, the latter of which inherited critical FSMO roles to ensure continuous directory service stability. Furthermore, the implementation of Nutanix Categories, Roles, and Projects enabled the application of granular authorization policies for the HR, Accounting, and Executive teams, which enforced the principle of least privilege where each department was restricted to the specific virtual assets and administrative permissions required for the roles. Additionally, the Nutanix Projects effectively isolated workloads and sandboxed infrastructure resources, which simplified administrative oversight and prevented conflicts with critical server operations and management tasks. Finally, the infrastructure was completed by the deployment of an IIS web server on G03SRV06, which hosted a variety of public and restricted internal web environments protected by Windows Authentication and network-level Internet Protocol (IP) restrictions.

Each section contained a numbered list which showed the steps. Sections' headers, which were bolded and centered, showed the actions taken. The bibliography appeared at the end of the report, followed by Appendix A, which contained the problem statement. Additional supporting documentation was included in subsequent appendices. The followings were the formatting keys:

buttons were bold, *options* were italicized, text entered into the computer was in Courier New, menu navigation was by the pipe symbol and italic words: *Start | Programs | MS Office | Word*.

Prism Central Authentication

The section explained the process of authentication and access to Nutanix Prism Central (PC) for the administrators inside the infrastructure. Microsoft Edge browser was used to perform login and administrative tasks on Nutanix PC, which served as the critical management hub for the infrastructure. Nutanix PC provided the centralized visibility and control necessary to perform administrative tasks. The following steps were documented on G03SRV03.

1. Clicked *Microsoft Edge* in *Windows Taskbar*
2. Pressed **Windows + T**
3. Typed `https://44.38.3.12:9440/ui/iam/login` in *Microsoft Edge*

Searchbar field

4. Pressed **Enter**
5. Typed `admin` in *Username* field
6. Inputted `[new secure password]` in *Password* field

G03SRV01VM Opening

The section detailed how the Virtual Machine console from the Nutanix Compute dashboard was started and logged in where the existing administrative credentials were used. Consistent access to the VM console was essential for work related to G03SRV01VM. The following steps were documented on G03SRV03.

1. Clicked *Microsoft Edge* in *Windows Taskbar*
2. Typed `https://44.38.3.12:9440/ui/iam/login` in *Microsoft Edge*

Searchbar

3. Pressed **Enter**
4. Clicked *Compute* dropdown | *VMs* | *G03SRV01VM* | *Launch Console* | **Send CtrlAltDel**
5. Typed [secure password] in *Password*
6. Pressed **Enter**

G03SRV03VM Opening

The section detailed how the VM console from the Nutanix Compute dashboard was started and logged in where the existing administrative credentials were used. Consistent access to the VM console was essential for work related to G03SRV03VM. The following steps were documented on G03SRV03.

1. Clicked *Microsoft Edge* in *Windows Taskbar*
2. Typed `https://44.38.3.12:9440/ui/iam/login` in *Microsoft Edge Searchbar*
3. Pressed **Enter**
4. Clicked *Compute* dropdown | *VMs* | *G03SRV03VM* | *Launch Console* | **Send CtrlAltDel**
5. Typed [secure password] in *Password*
6. Pressed **Enter**

OpenLDAP GPO Setup on G03SRV03VM

The section detailed the modification of the Default Domain Policy to adjust Lightweight Directory Access Protocol security requirements for the enterprise environment. The Group Policy Object (GPO) configurations were vital to the business case as the configurations ensured seamless interoperability between Windows Domain Controllers and Linux-based OpenLDAP services, which allowed for a unified authentication strategy across diverse platforms. **Refer to Appendix B: System and Network Configuration.**

1. Clicked *Server Manager* in *Windows Taskbar* window | *Tools* | *Group Policy Management*
2. Right-clicked *Default Domain Policy* under *c242-03-a.group03.c24200.cit.lcl* field
3. Clicked *Edit...* | *Computer Configuration* | *Policies* | *Windows Settings* | *Security Settings* | *Local Policies* | *Security Options*
4. Double-clicked *Domain controller: LDAP server channel binding token requirements*
5. Selected *Define this policy setting* | *When supporting* | **OK** in *Domain controller: LDAP server channel binding token requirements* window
6. Double-clicked *Domain Controller: LDAP server signing requirements*
7. Selected *Define this policy* | *None* | **OK** in *Domain Controller: LDAP server signing requirements* window
8. Double-clicked *Domain Controller: LDAP server signing requirements Enforcements*
9. Selected *Define this policy setting* | *Disabled* | **OK** in *Domain Controller: LDAP server signing requirements Enforcements* window
10. Double-clicked *Network security: LDAP client encryption requirements*

11. Selected *Define this policy setting* | *negotiate sealing* | **OK** in *Network security: LDAP client encryption requirements* window
12. Double-clicked *Network security: LDAP client signing requirements*
13. Selected *Define this policy setting* | *Negotiate signing* | **OK** in *Network security: LDAP client signing requirements* window
14. Clicked *Close* in *Group Policy Management Editor* window

OpenLDAP GPO Setup on G03SRV01VM

The section detailed the modification of the Default Domain Policy to adjust LDAP security requirements for the enterprise environment. The GPO configurations were vital to the business case as the configurations ensured seamless interoperability between Windows Domain Controllers and Linux-based OpenLDAP services, which allowed for a unified authentication strategy across diverse platforms. **Refer to Appendix B: System and Network Configuration.**

1. Clicked *Server Manager* in *Windows Taskbar* window | *Tools* | *Group Policy Management*
2. Right-clicked *Default Domain Policy* under *c242-03-a.group03.c24200.cit.lcl*
3. Clicked *Edit...* | *Computer Configuration* | *Policies* | *Windows Settings* | *Security Settings* | *Local Policies* | *Security Options*
4. Double-clicked *Domain controller: LDAP server channel binding token requirements*
5. Selected *Define this policy setting* | *When supporting* | **OK** in *Domain controller: LDAP server channel binding token requirements* window
6. Double-clicked *Domain Controller: LDAP server signing requirements*
7. Selected *Define this policy* | *None* | **OK** in *Domain Controller: LDAP server signing requirements* window

8. Double-clicked *Domain Controller: LDAP server signing requirements Enforcements*
9. Selected *Define this policy setting | Disabled | OK* in *Domain Controller: LDAP server signing requirements Enforcements* window
10. Double-clicked *Network security: LDAP client encryption requirements*
11. Selected *Define this policy setting | negotiate sealing | OK* in *Network security: LDAP client encryption requirements* window
12. Double-clicked *Network security: LDAP client signing requirements*
13. Selected *Define this policy setting | Negotiate signing | OK* in *Network security: LDAP client signing requirements* window
14. Clicked *Close* in *Group Policy Management Editor* window

Active Directory Domains Integration on Nutanix PC

The section detailed the integration of Nutanix Prism Central with the Group03 Active Directory (AD) domains via the Identity and Access Management (IAM) system. The establishment of the connection enabled Centralized Authentication for administrators to manage infrastructure where the existing credentials could be used, which omitted the need for distinct local accounts. The integration streamlined user lifecycle management, enhanced security audit, and ensured uniform access control throughout the enterprise virtualization platform. **Refer to Appendix B: System and Network Configuration.**

1. Clicked *Prism Central Setting | Authentication* under *Users and Roles | Manage* in *Admin Center* window | **Add Identity Provider** | *Active Directory/OpenLDAP*
2. Typed G03SRV01LDAP in *Name* field
3. Inputted group03.c24200.cit.lcl in *Domain* field
4. Typed ldap://44.38.3.31:389 in *Directory URL* field

5. Inputted administrator@group03.c24200.cit.lcl in *Username* field
6. Typed [Group 03 administrator password] in *Password* field
7. Clicked **Save | Add Identity Provider | Active Directory/OpenLDAP**
8. Typed G03SRV03LDAP in *Name* field
9. Inputted c242-03-a.group03.c24200.cit.lcl in *Domain* field
10. Typed ldap://44.38.3.20:389 in *Directory URL* field
11. Inputted administrator@c242-03-a.group03.c24200.cit.lcl in *Username* field
12. Typed [Group 03 A administrator password] in *Password* field

TrueNAS ISO Creation on Nutanix PC

The section detailed how TrueNAS installation media was imported into the Nutanix environment from a centralized network repository. The process established the foundational software image required to deploy high-availability Network Attached Storage (NAS) across the enterprise. As a mapped network drive was employed for image acquisition, the organization guaranteed all infrastructure teams were deployed from a validated, uniform version of the storage operating system. The practice mitigated installation errors and preserved uniformity across the datacenter. **Refer to Appendix B: System and Network Configuration.**

1. Clicked *Compute* dropdown | *Images* | **Add Image** | **+ Add File**
2. Right-clicked *This PC*
3. Clicked *Map network drive*
4. Typed \\rtfm.cit.lcl in *Folder* field
5. Clicked **Browse...** | *rtfm.cit.lcl*
6. Typed CIT\[school account username] in *User Name* field

7. Inputted [school account password] in *Password* field
8. Clicked *rtfm.cit.lcl* | *Pub* | *CNIT24200* | **OK** in *Browse For Folder* window | **Finish** | *TrueNAS-13.0-U6.2* in *Open* window | **Open** in *Open* window | **Next** in *Select Image* window | **Save** in *Select Location* window

TrueNAS VM Deployment on Nutanix PC

The section detailed the provision of a high-capacity virtual machine on the Nutanix platform to host the TrueNAS operating system. The deployment was a fundamental requirement for the business case as it established a robust solution, which enabled centralized file share and data redundancy for the enterprise. As specific hardware resources were allocated such as 8-gigabyte (GB) of memory and multi-disk configurations, the organization ensured the storage server could handle concurrent data requests across the network. **Refer to Appendix B: System and Network Configuration.**

1. Clicked *Compute* | *VMs* | **Create VM**
2. Typed G03TRUENAS in *Name* field
3. Clicked *Cluster* dropdown | *G03NUTXSRV01* | *Advanced Settings*
4. Typed 8 in *Memory* field
5. Inputted 2 in *Cores Per CPU* field
6. Selected *Enable Memory Overcommit*
7. Clicked **Next** in Configuration window | **Attach Disk** | *Type* dropdown | *CD-ROM* | *Operation* dropdown | *Clone from Image* | *Image* dropdown | *TrueNAS-13.0-U6.2.iso* | **Save** in *Attach Disk* window | *Attached Disk* | *Storage Container* dropdown | *G03VM*
8. Typed 32 in *Capacity* field
9. Clicked **Save** in *Attach Disk* window

10. Selected *Attach Disk*
11. Typed 250 in *Capacity* field
12. Clicked **Save** in *Attach Disk* window
13. Repeated steps 11-13
14. Clicked **Attach to Subnet** | **Save** in *Attach to Subnet* window | **Next** in *Resources* window | *Timezone* dropdown | *(UTC - 04:00) America/Indianapolis* | **Next** in *Management* window | **Create VM**

TrueNAS Installation on Nutanix PC

The section detailed the installation of the TrueNAS operating system onto the virtualized hardware environment. The process was critical to the business case which transformed the provisioned virtual machine into a functional storage appliance, which established the root security credentials and Unified Extensible Firmware Interface (UEFI) boot parameters necessary for a stable production environment. After successful installation and installation media unmounting processes were completed, the organization moved from the provisioning phase to an operational state, ready to serve as the backbone for enterprise data management.

Refer to Appendix B: System and Network Configuration.

1. Clicked *Compute* | *VMs* | *G03TRUENAS* | *More* dropdown | *Power On* | *Launch Console*
2. Selected *Install/Upgrade* | **OK** in *TrueNAS 13.0-U6.2 Console Setup* window | *da0*
Nutanix VDISK – 32.0 GiB | **OK** in *Choose destination media* window | **Yes** in *TrueNAS installation* window
3. Typed [new secure password] in *Password* field
4. Inputted [new secure password] in *Confirm Password* field

5. Selected **OK** in *Enter your root password; cancel for no root password* window | **Boot via UEFI** | **OK** in *The TrueNAS installation on da0 succeeded!* window | *Shutdown System* | **OK** in *TrueNAS 13.0-U6.2 Console Setup* window
6. Clicked **Unmount ISO** in *G03TRUENAS - Profile 1 - Microsoft Edge* window | **Unmount**
7. Clicked *Close* in *G03TRUENAS - Profile 1 - Microsoft Edge* window
8. Clicked *VMs* in *Prism Central* window | *G03TRUENAS* | *More* dropdown | *Power On*

G03TRUENAS Opening

The section detailed how the VM console was launched from the Nutanix Compute dashboard. Consistent access to the VM console was necessary for tasks involved with G03TRUENAS. The following steps were documented on G03SRV03.

1. Clicked *Microsoft Edge* in *Windows Taskbar*
2. Pressed **Ctrl + T** in *Microsoft Edge* window
3. Typed `https://44.38.3.12:9440/ui/iam/login` in *Microsoft Edge Searchbar* field
4. Pressed **Enter**
5. Clicked *Compute* dropdown | *VMs* | *G03TRUENAS* | *Launch Console* | *Windows Screen*

Prism Element Authentication on G03NUTXSRV01

The section detailed the authentication process to access the single cluster management dashboard, which was Nutanix Prism Element (PE). Consistent access to the VM console was necessary for tasks which involved Nutanix PE. PE was critical to the business case, which provided direct, localized management of the G03NUTXSRV01 cluster, and allowed for low-level configuration of storage, network communication, and hardware resources specific to the physical host environment. Microsoft Edge browser was used to perform login and administrative tasks on Nutanix PE. **Refer to Appendix B: System and Network**

Configuration.

1. Clicked *Microsoft Edge* in *Windows Taskbar*
2. Typed `https://44.38.3.40:9440/console` in *Microsoft Edge Searchbar* field
3. Pressed **Enter**
4. Typed `admin` in *username* field
5. Inputted `[new secure password]` in *password* field
6. Pressed **Enter**

TrueNas IP Configuration on G03TRUENAS

The section detailed the manual assignment of a static IP version 4 (IPv4) address to the TrueNAS management interface. The configuration, which was essential for the business case, ensured a predictable and permanent network location (44.38.3.50) for the enterprise storage server. The static IP address was implemented instead of Dynamic Host Configuration Protocol (DHCP), which guaranteed all connected clients and administrative tools maintain uninterrupted access to the storage volumes and management dashboard. **Refer to Appendix B: System and Network Configuration.**

1. Typed 1 into *Enter an option from 1-11*
2. Pressed **Enter**
3. Typed 1 into *Select an interface (q to quit):*
4. Pressed **Enter**
5. Typed n into *Remove the current settings of this interface?*
6. Pressed **Enter**
7. Typed n into *Configure interface for DHCP*
8. Pressed **Enter**
9. Typed y into *Configure IPv4*
10. Pressed **Enter** | **Enter** in *Interface name* window
11. Typed 44.38.3.50/24 into *IPv4 Address*
12. Pressed **Enter**
13. Typed n into *IPv6 Address*
14. Pressed **Enter**

TrueNas DNS Configuration on G03TRUENAS

The section detailed the configuration of Domain Name System settings on the TrueNAS appliance, which focus on link establishment to the enterprise naming environment. The process was mandatory for an AD domain registration. As the TrueNAS system was pointed to the primary domain controller (44.38.3.31) and the group03.c24200.cit.lcl namespace was defined, the server could successfully locate and authenticate with the Domain Controller, which allowed storage permissions and user access to be managed centrally. **Refer to Appendix B: System and Network Configuration.**

1. Typed 6 in *Enter an option from 1-11* field

2. Pressed **Enter**
3. Typed `group03.c24200.cit.lcl` in *DNS Domain [local]* field
4. Pressed **Enter**
5. Typed `44.38.3.31` in *DNS Nameserver 1* field
6. Pressed **Enter** | **Enter** in *DNS Nameserver 2* window

G03TRUENAS Password Setup

The section detailed the final security step for the TrueNAS web interface, which involved establishment of the root account credentials for remote management and the principal administrative access point for the enterprise storage system. Through the implementation of a robust, synchronized password for the web graphical user interface, the organization restricted access to high-level storage configurations to authorized individuals, hence, safeguarded sensitive company data from both unauthorized access and inadvertent misconfiguration. The following steps were documented on G03SRV03.

1. Clicked *Microsoft Edge* in *Windows Taskbar*
2. Typed `https://44.38.3.50` in *Microsoft Edge Searchbar* field
3. Pressed **Enter**
4. Typed `[new secure password]` under *set new root account password*
5. Inputted `[new secure password]` under *Confirm Password*
6. Clicked **SIGN IN**

G03TRUENAS Web Interface Opening

The section detailed the initial login procedure for the TrueNAS web-based graphical user interface, facilitated through a secure browser connection. The process moved administrative tasks from the command-line interface to a more user-friendly management dashboard. Access to the web interface was essential for advanced storage configurations, which included RAID pools creation, user shares setup, and system health monitor. The features were crucial for the enterprise's data availability and security maintenance. Consistent access to the TrueNAS web interface was important for tasks involved with G03TRUENAS. The following steps were performed on G03SRV03.

1. Clicked *Microsoft Edge* in *Windows Taskbar*
2. Typed `https://44.38.3.50` in *Microsoft Edge Searchbar*
3. Pressed **Enter**
4. Typed `root` in *username*
5. Inputted `[new secure password]` in *password* field
6. Pressed **Enter**

Storage Pool Creation and Active Directory Registration in G03TRUENAS Web Interface

The section detailed the final procedures for the TrueNAS configuration as a robust, enterprise-level storage solution. The process accomplished two fundamental objectives: Unified Identity Management and Data Redundancy. Through the registration of the appliance with the group03.c24200.cit.lcl AD domain, the organization facilitated streamlined permissions management where pre-existing domain accounts were leveraged. Concurrently, the establishment of the Group03NAS storage pool, which utilized multiple virtual disks, guaranteed

the enterprise a dependable, high-performance volume for the storage of essential departmental data. **Refer to Appendix B: System and Network Configuration.**

1. Clicked *Directory Service | Active Directory*
2. Typed `group03.c24200.cit.lcl` in *Domain Name** field
3. Inputted Administrator in *Domain Account Name** field
4. Typed [Administrator's secure password] in *Domain Account Password* field
5. Clicked **SAVE** | *Storage | Pools* | **ADD** in *Pools* window | **CREATE POOL** |
6. Typed Group03NAS in *Name* field
7. Selected *da1* in *Available Disks* window | *da2* in *Available Disks* window
8. Clicked **Right Arrow** under *Data VDevs* | **CREATE**
9. Selected *Confirm* in Warning page window
10. Clicked **CREATE POOL**

Datasets Creation in TrueNAS

The section detailed how the G03NAS storage pool was logically divided into specific Datasets. The setup created a structured file system, which allowed the organization to separate regular user data from important recovery assets. As datasets were used instead of regular folders, the administration could set specific quotas, compression settings, and snapshot policies for each department. The method ensured efficient use of storage resources and easier maintenance with clear data boundaries across the organization. **Refer to Appendix B: System and Network Configuration.**

1. Clicked *Menu | Storage | Pools | G03NAS (System Dataset Pool) | Dataset Actions | Add Dataset*

2. Typed `GeneralSet03` in *Name* field
3. Clicked **SUBMIT** | *G03NAS (System Dataset Pool)* | *Dataset Actions* in *G03NAS* window | *Add Dataset*
4. Typed `BackupSet03` in *Name* field
5. Clicked **SUBMIT**

SMB Share and Access Control Lists Configurations for BackupSet03

The section detailed the deployment of a Windows-compatible network share for the BackupSet03 dataset, which utilized the SMB protocol. The process provided the mechanism for Windows-based clients and servers to securely transmit backup data to the TrueNAS appliance. As Access Control Lists (ACLs) was applied with a "Restricted" preset and the GROUP03\administrator group was granted permissions, the organization enforced a high-security posture, which ensured sensitive backup archives were only accessible to authorized administrative personnel and prevented unauthorized data tampering or exfiltration. The following steps were implemented on the TrueNAS web interface. **Refer to Appendix B: System and Network Configuration.**

1. Clicked *Sharing* | *Windows Share (SMB)* | **ADD**
2. Typed `/mnt/G03NAS/BackupSet03` in *Path* field
3. Clicked **SUBMIT** | **ENABLE SERVICE** | **CLOSE** in *SMB Service* window | **CONFIGURE NOW** in *Configure ACL* window |
4. Selected *Select a preset ACL* | *Default ACL Options* dropdown | *RESTRICTED* | **ADD ACL ITEM** | *Who** dropdown | *Group* | *Group** dropdown | *GROUP03\administrator*
5. Clicked **SAVE**

Share Configuration with SMB and ACLs for GeneralSet03

The section detailed how the main network share for the GeneralSet03 dataset was set up on TrueNAS web interface. The share served as a central location for general organizational data. The process allowed for Collaborative Data Access across the domain. Unlike the backup shares, which had limited access, the SMB share was configured to work with Active Directory's GROUP03 AD domain users, which ensured all authenticated employees had a specific space for file storage and share. As the "Restricted" ACL preset and the domain users group were added, the organization maintained a "Secure-by-Default" approach and still allowed the necessary connectivity for everyday business activities. **Refer to Appendix B: System and Network Configuration.**

1. Clicked *Sharing | Windows Share (SMB) | ADD*
2. Typed `/mnt/G03NAS/GeneralSet03` in *Path* field
3. Clicked **SUBMIT | ENABLE SERVICE | CLOSE** in *SMB Service* window |
CONFIGURE NOW in *Configure ACL* window |
4. Selected *Select a preset ACL | Default ACL Options* dropdown | *RESTRICTED* | **ADD**
ACL ITEM | *Who** dropdown | *Group* | *Group** dropdown | *GROUP03\domain users*
5. Clicked **SAVE**

Microsoft Backup Implementation on G03SRV01VM

The section detailed the installation and backup schedule implementation of the built-in Windows Server Backup feature on G03SRV01VM. The configuration was a key part of the business case's disaster recovery plan, which ensured important data, specifically the System State like the Active Directory database, registry, boot files, and user-specific directories, was backed up daily. As remote TrueNAS SMB share (\\44.38.3.50\BackupSet03) was used, the organization achieved off-box redundancy, which protected the server's identity and configuration from hardware or VM-level failures. **Refer to Appendix B: System and Network Configuration.**

1. Right-clicked *PowerShell* on *Windows Taskbar*
2. Clicked *Run as administrator*
3. Typed `Install-WindowsFeature Windows-Server-Backup`
4. Pressed **Enter**
5. Clicked *Tools* | *Windows Server Backup* | *Local Backup* | *Backup Schedule* under *Actions* tab | **Next** in *Getting Started* window | *Custom* | **Next** in *Select Backup Configuration* window | **Add Items**
6. Selected *System state* | *Users* under *Local disk (C:)*
7. Clicked **OK** in *Select Items* window | **Next** in *Select Items for Backup* window | *Select time of day* dropdown | *5:00AM* | **Next** in *Specify Backup Time* window | *Back up to a shared network folder* | **Next** in *Specify Destination Type* window
8. Selected **OK** in *Windows Server Backup* window
9. Typed `\\44.38.3.50\BackupSet03` in *Location* field
10. Clicked **Next** in *Specify Remote Shared Folder* window

11. Typed Administrator in *User name* field
12. Inputted [Administrator's password] in *Password* field
13. Clicked **OK** in *Register Backup Schedule* window | **Finish** in *Confirmation* window | **Close** in *Summary* window | *Close* in *wbadmin - [Windows Server Backup (Local)\Local Backup]* window

Backup Rename and Deletion Script Creation on G03SRV01VM

The section detailed the creation of specialized batch scripts (*renameBackup.cmd* and *deleteBackup.cmd*), which were designed to manage the rename process and retention of server backups. The process was vital because standard Windows Server Backups to network shares could overwrite previous backup versions, which required manual intervention to organize. As the rename and deletion process were automated within the *C:\mainScripts* directory, the organization ensured a clean, rotating archive of backups on the TrueNAS storage. The programmatic approach prevented the storage pool from resource exhaustion and maintained a history of recoverable points for the domain controller. **Refer to Appendix B: System and Network Configuration. See Appendix C: Backup Script Configuration for script.**

1. Clicked *File Explorer* in *Windows Taskbar* | *This PC* | *Local Disk (C:)* in *Devices and drives* window | *New* in *toolbar* window | *Folder*
2. Typed *mainScripts* in *Name* field
3. Clicked *Close* in *File Explorer* window | *Notepad* in *Windows Taskbar*
4. Typed *renameBackup* script as seen in Appendix C
5. Pressed **Ctrl+S** to save script to destination
6. Typed *renameBackup.cmd* in *File name* field

7. Clicked *File Explorer* in *Windows Taskbar* | *This PC* | *Local Disk (C:)* in *Devices and drives* | *mainScripts* | **Save** | *Notepad* in *Windows Taskbar*
8. Typed `deleteBackup` script as seen in Appendix C
9. Pressed **Ctrl+S** to save script to destination
10. Typed `deleteBackup.cmd` in *File name* field
11. Clicked *This PC* | *Local Disk (C:)* in *Devices and drives* window | *mainScripts* | **Save**

Task Scheduler Configuration for Backup Scripts on G03SRV01VM

The section detailed the configuration of the Windows Task Scheduler to automate the execution of the maintenance scripts. The process ensured the "Rename" and "Delete" logic occurs consistently every morning at 6:30 AM, shortly after the primary backup window finished. As the manual scripts were transitioned into scheduled tasks, the administration created an automated maintenance cycle, which guaranteed the backup repository remained organized and up-to-date without daily human oversight. The automated scripts significantly improved the reliability of the enterprise's recovery posture. **Refer to Appendix B: System and Network Configuration.**

1. Clicked *Server Manager* in *Windows Taskbar* | *Tools* | *Task Scheduler* | *Action* tab | *Create Basic Task...* in *Action* dropdown
2. Typed `delBackup` in *Name* field
3. Clicked **Next** in *Create a Basic Task* window | *Daily* under *When do you want the task to start?* field | **Next** in *Trigger* window
4. Inputted `6:30:00 AM` in *Start* field

5. Clicked **Next** in *Daily* window | *Start a program* under *What action do you want the task to perform?* field | **Next** in *Action* window | *Browse...* in *Program/script* window | *This PC* | *Local Disk (C:)* | *mainScripts* | *deleteBackup*
6. Selected **Open** in *Open* window | **Next** in *Start a Program* window | **Finish** in *Summary* window
7. Clicked *Create Basic Task...* in *Actions* window
8. Typed `renameBackup` in *Name* field
9. Clicked **Next** in *Create a Basic Task* window | *Daily* in *When do you want the task to start?* window | **Next** in *Trigger* window
10. Inputted `6:30:00 AM` in *Start* field
11. Clicked **Next** in *Daily* window | *Start a program* in *What action do you want the task to perform?* window | **Next** in *Action* window | *Browse...* in *Program/script* window | *This PC* | *Local Disk (C:)* | *mainScripts* | *renameBackup*
12. Selected **Open** in *Open* window | **Next** in *Start a Program* window | **Finish** in *Summary* window | *Close* in *Task Scheduler* window

New Subnets Configuration on Nutanix PC

The section detailed the creation of the LAN1903 subnet across the Nutanix clusters. The new subnet was created to provide a dedicated communication layer for the project's virtual assets. Nutanix IP Address Management (IPAM) was used to automatically assign network addresses, which reduced the risk of IP address conflicts. As a specific VLAN ID (1903) and default gateway (44.38.103.1) were defined, the organization ensured traffic was properly segmented and could be routed correctly. Furthermore, the enterprise DNS server (44.38.3.31) was configured on the subnet, which guaranteed all newly provisioned VMs could immediately

resolve domain resources and join the Active Directory environment. **Refer to Appendix B: System and Network Configuration.**

1. Clicked *Network & Security* | *Subnets* | **Create Subnet**
2. Typed LAN1903 in *Name* field
3. Inputted 1903 in *VLAN ID* field
4. Clicked *Cluster* dropdown | *G03NUTXSRV02* | *IP Assignment Service* dropdown |
Nutanix IPAM
5. Typed 44.38.103.0/24 in *Network IP Address / Prefix*
6. Inputted 44.38.103.1 in *Gateway IP Address* field
7. Typed 44.38.3.31 in *Domain Name Servers* field
8. Clicked **Create** in *Create Subnet* window
9. Selected **Create Subnet**
10. Repeated steps 3-4
11. Clicked *Cluster* dropdown | *G03NUTXSRV01* | *IP Assignment Service* dropdown |
Nutanix IPAM
12. Repeated steps 6-9

G03SRV04 Deployment on Nutanix PC

The section detailed the deployment of G03SRV04 from the G03WIN2025TEMP template. Deployment from a template ensured a standardized configuration across VMs in the environment and reduced setup time. G03SRV04 deployment on Nutanix PC provided a centralized management platform, which allowed for efficient resource allocation and simplified oversight of the server's lifecycle required for G03SRV04's future role as a critical Domain Controller. As a future Domain Controller, G03SRV04 would provide the high-availability

authentication and redundant directory services required to prevent enterprise-wide downtime.

Refer to Appendix B: System and Network Configuration.

1. Clicked *Compute* | *VMs* | **Create VM from Template** | *VM Template* dropdown | *G03WIN2025TEMP* | **Begin**
2. Typed G03SRV04 in *Name* field
3. Clicked *Cluster* dropdown | *G03NUTXSRV01* | **Advanced Deploy** | **Next** in *Configuration* window | *Delete icon* in *G03NUTXSRV02* window | **Attach to Subnet** | *Subnet* dropdown | **LAN1903**
4. Typed 44.38.103.2 in *IP Address* field
5. Clicked **Save** in *Attach to Subnet* window | **Next** in *Resources* window | **Next** in *Management* window | **Deploy** in *Review* window | *G03SRV04* | **More** | *Power On*
6. Pressed **Ctrl + R**
7. Clicked **Launch Console** | **Send CtrlAltDel** in *G03SRV04 - Profile 1 - Microsoft Edge* window
8. Typed [Administrator's default password] in *Administrator* field
9. Pressed **Enter**

G03SRV04 Opening

The section detailed how the VM console from the Nutanix Compute dashboard was started and logged in where the existing administrative credentials were used. Consistent access to the VM console was necessary for tasks involved with G03SRV04. The following steps were documented on G03SRV03.

1. Clicked *Microsoft Edge* in *Windows Taskbar*

2. Typed `https://44.38.3.12:9440/ui/iam/login` in *Microsoft Edge Searchbar*
3. Pressed **Enter**
4. Clicked *Compute* dropdown | *VMs* | *G03SRV04* | *Launch Console* | **Send CtrlAltDel**
5. Typed [secure password] in *Password*
6. Pressed **Enter**

G03SRV04 Network Configuration

The section detailed how IP version 6 (IPv6) was disabled, static IPv4 address (44.38.103.2) was assigned, and DNS was set to G03SRV04. IP addresses, which functioned as unique numerical identifiers, enabled precise and reliable network communication between specific nodes within a network architecture. Proper IP configuration allowed the new server to communicate with existing domain controllers and services. **Refer to Appendix B: System and Network Configuration.**

1. Clicked *Server Manager* in *Windows Taskbar* | *Local Server* in *Server Manager* window | *IPv4 address assigned by DHCP, IPv6 enabled*
2. Right-clicked *Ethernet* | *Properties*
3. Unselected *Internet Protocol Version 6 (TCP/IPv6)*
4. Double-clicked *Internet Protocol Version 4 (TCP/IPv4)*
5. Selected *Use the following IP address* | *Use the following DNS server addresses*
6. Typed `44.38.103.2` in *IP Address* field
7. Inputted `255.255.255.0` in *Subnet mask* field
8. Typed `44.38.103.1` in *Default gateway* field
9. Inputted `44.38.3.31` in *Preferred DNS server* field

10. Clicked **OK** in *Internet Protocol Version 4 (TCP/IPv4) Properties* window | **OK** in *Ethernet Properties* window | *Close* in *Network Connections* window

G03SRV04 Computer Name and AD Domain Configurations

The section detailed how G03SRV04 was enrolled from a workgroup to the group03.c24200.cit.lcl domain. The process allowed the server to participate in domain-based security, Group Policy, and centralized management. Domain enrollment, which provided the centralized security and management necessity, ensured consistent user access and organizational control over all network assets. **Refer to Appendix B: System and Network Configuration.**

1. Opened G03SRV04 (see section)
2. Clicked *Server Manager* in *Windows Taskbar* | *Local Server* in *Server Manager* window | *G03WIN2025TEMP* in *Computer name* window
3. Typed G03SRV04 in *Computer description* field
4. Clicked **OK** in *System Properties* window
5. Typed G03SRV04 in *Computer name* field
6. Inputted group03.c24200.cit.lcl in *Domain* field | **OK** in *Computer Name/Domain Changes*
7. Typed Administrator in *User name* field
8. Inputted [Group03 Administrator's password] in *Password* field
9. Clicked **OK** in *Windows Security* window | **OK** in *Computer Name/Domain Changes* window | **OK** in *You must restart your computer to apply these changes* window | **Close** in *System Properties* window | **Restart Now** in *Microsoft Windows* window

G03SRV04 Domain Controller Promotion

The section detailed the promotion of G03SRV04 to an AD Domain Controller (DC) within the group03.c24200.cit.lcl domain. The objective was completed through navigation within the features of Active Directory server manager, and the use of the parent DC administrator credentials during installation, so the prerequisite checks can run successfully. Additional DC provides essential redundancy and load balancing to ensure continuous network authentication and service availability. The following steps were documented on G03SRV03.

1. Opened Prism Central (see section)
2. Clicked *Compute* | *VMs* | *G03SRV04* | **Launch Console** | **Send CtrlAltDel** in *G03SRV04 - Profile 1 - Microsoft Edge* window | *Other user*
3. Typed Group03\Administrator in *User name* field
4. Inputted [Group03 Administrator's password] in *Password* field
5. Pressed **Enter**
6. Clicked *Server Manager* in *Windows Taskbar* window | *Manage* | *Add Roles and Features* | **Next** in *Before You Begin* window | **Next** in *Installation Type* window | **Next** in *Server Selection* window
7. Selected *Active Directory Domain Services* | **Add Features** in *Add Roles and Features Wizard* window
8. Clicked **Next** in *Server Roles* window | **Next** in *Features* window | **Next** in *AD DS* window | **Install** | **Close** in *Results* window | **Notifications (icon)** | *Promote this server to a domain controller* | **Next** in *Deployment Configuration* window
9. Typed [new secure password] in *Password* field
10. Inputted [new secure password] in *Confirm password* field

11. Clicked **Next** in *Domain Controller Options* window | **Next** in *DNS Options* window | **Next** in *Additional Options* | **Next** in *Paths* window | **Next** in *Review Options* window | **Install** | **Close** in *You're about to Sign out* window

AD Subnet and Site Creation on G03SRV04

The section detailed the configuration of logical network boundaries within the AD environment. As the GROUP03SITE was established and the 44.38.103.0/24 subnet was defined, the enterprise could control replication topology and ensured the client traffic was routed efficiently to local resources.

1. Clicked *Server Manager* in *Windows Taskbar* window | *Tools* | *Active Directory Site and Services*
2. Right-clicked *Sites*
3. Clicked *New Site...*
4. Typed GROUP03SITE in *Name* field
5. Clicked *DEFAULTIPSITELINK* | **OK** in *New Object - Site* window | **OK** in *Active Directory Domain Services* window
6. Right-clicked *Subnets* | *New Subnet...*
7. Typed 44.38.103.0/24 in *Prefix* field
8. Clicked *GROUP03SITE* | **OK** in *New Object - Subnet* window

G03S2WIN11 Deployment on Nutanix PC

The section detailed the deployment of G03S2WIN11 from the G03WIN11TEMP template. Deployment from a template ensured a standardized configuration across VMs in the environment and reduced setup time. G03S2WIN11 provided a secure, standardized workstation environment, which ensured employees have reliable access to the enterprise tools and resources necessary for daily business operations. **Refer to Appendix B: System and Network Configuration.**

1. Clicked *Compute* | *VMs* | **Create VM from Template** | *VM Template* dropdown | *G03WIN11TEMP* | **Begin**
2. Typed G03S2WIN11 in *Name* field
3. Clicked *Cluster* dropdown | *G03NUTXSRV01* | **Advanced Deploy** | **Next** in *Configuration* window | *Delete icon* in *G03NUTXSRV02* window | **Attach to Subnet** | *Subnet* dropdown | **LAN1903**
4. Typed 44.38.103.3 in *IP Address* field
5. Clicked **Save** in *Attach to Subnet* window | **Next** in *Resources* window | **Next** in *Management* window | **Deploy** in *Review* window | *G03S2WIN11* | **More** | *Power On*
6. Pressed **Ctrl + R**
7. Clicked **Launch Console** | **Send CtrlAltDel** in *G03S2WIN11 - Profile 1 - Microsoft Edge* window
8. Typed [WindowsTemp's default password] in *WindowsTemp* field
9. Pressed **Enter**

G03S2WIN11 Opening

The section detailed how the VM console from the Nutanix Compute dashboard was started and logged in where the existing administrative credentials were used. Consistent access to the VM console was necessary for tasks involved with G03S2WIN11. The following steps were documented on G03SRV03.

1. Typed `https://44.38.3.12:9440/ui/iam/login` in *Microsoft Edge*
Searchbar
2. Pressed **Enter**
3. Clicked *Compute* dropdown | *VMs* | *G03S2WIN11* | *Launch Console* | **Send CtrlAltDel**
4. Typed [secure password] in *Password*
5. Pressed **Enter**

G03S2WIN11 Computer Name and IP Configurations

The section detailed the manual configuration of the IPv4 stack, which included setting static IP to 44.38.103.3 and setting preferred DNS to the virtual DC at 44.38.3.20. The IP configurations were necessary for G03S2WIN11 to locate the DC and successfully join the AD domain. **Refer to Appendix B: System and Network Configuration.**

1. Right-clicked **Start**
2. Clicked *Settings* | *Network & internet* | *Ethernet* | **Edit** in *IP assignment* window | *Edit IP settings* dropdown | *Manual*
3. Selected *IPv4*
4. Typed `44.38.103.3` in *IP address* field
5. Inputted `255.255.255.0` in *Subnet mask* field
6. Typed `44.38.103.1` in *Gateway* field

7. Inputted 44.38.3.31 in *Preferred DNS* field
8. Clicked **Save** in *Edit IP settings* window | *System* | *Rename*
9. Typed G03S2WIN11 in *Current PC name: G03WIN11TEMP* field
10. Clicked **Next** in *Rename your PC* window | **Restart now** in *Rename your PC* window

G03S2WIN11 AD Domain Configuration

The section detailed how the computer was renamed and authenticated with AD domain administrator credentials within group03.c24200.cit.lcl domain, which enrolled the machine to group03.c24200.cit.lcl domain. Successful AD domain enrollment allowed the client to receive AD domain Group Policy Objects and authenticate. **Refer to Appendix B: System and Network Configuration.**

1. Right-clicked **Start**
2. Clicked *Settings* | *System* | *About* | *Advanced System Settings* | *Computer Name* | **Change...**
3. Typed group03.c24200.cit.lcl in *Domain* field
4. Clicked **OK** in *Computer Name/Domain Changes*
5. Typed Administrator in *User name* field
6. Inputted [Group03 Administrator's password] in *Password* field
7. Clicked **OK** in *Computer Name/Domain Changes* | **OK** in *Computer Name/Domain Changes* | **Close** in *System Properties* window | **Restart Now** in *Microsoft Windows* window

G03SRV05 Deployment on Nutanix PC

The section detailed the deployment of G03SRV05 from the G03WIN2025TEMP template. Deployment from a template ensured a standardized configuration across VMs in the environment and reduced setup time. G03SRV05 deployment on Nutanix PC provided a centralized management platform, which allowed for efficient resource allocation and simplified oversight of the server's lifecycle required for G03SRV05's future role as a critical Domain Controller. As a future Domain Controller, G03SRV05 would provide the high-availability authentication and redundant directory services required to prevent enterprise-wide downtime.

Refer to Appendix B: System and Network Configuration.

1. Clicked *Compute* | *VMs* | **Create VM from Template** | *VM Template* dropdown | *G03WIN2025TEMP* | **Begin**
2. Typed G03SRV05 in *Name* field
3. Clicked *Cluster* dropdown | *G03NUTXSRV01* | **Advanced Deploy** | **Next** in *Configuration* window | *Delete icon* in *G03NUTXSRV02* window | **Attach to Subnet** | *Subnet* dropdown | **G03NUTXSRV01**
4. Selected **Save** in *Attach to Subnet* window | **Next** in *Resources* window | **Next** in *Management* window | **Deploy** in *Review* window | *G03SRV05* | **More** | *Power On*
5. Pressed **Ctrl + R**
6. Clicked **Launch Console** | **Send CtrlAltDel** in *G03SRV05 - Profile 1 - Microsoft Edge* window
7. Typed [Administrator's default password] in *Administrator* field
8. Pressed **Enter**

G03SRV05 Opening

The section detailed how the VM console from the Nutanix Compute dashboard was started and logged in where the existing administrative credentials were used. Consistent access to the VM console was necessary for tasks involved with G03SRV05. The following steps were documented on G03SRV03.

1. Typed `https://44.38.3.12:9440/ui/iam/login` in *Microsoft Edge Searchbar*
2. Pressed **Enter**
3. Clicked *Compute* dropdown | *VMs* | *G03SRV05* | *Launch Console* | **Send CtrlAltDel**
4. Typed [secure password] in *Password*
5. Pressed **Enter**

G03SRV05 Network Configuration

The section detailed how IPv6 was disabled, static IPv4 address (44.38.3.67) was assigned, and DNS was set to G03SRV05. IP addresses, which functioned as unique numerical identifiers, enabled precise routing and reliable communication between specific nodes within a network architecture. Proper IP configuration allowed the new server to communicate with existing domain controllers and services. **Refer to Appendix B: System and Network Configuration.**

1. Opened G03SRV05 (see section)
2. Clicked *Server Manager* in *Windows Taskbar* window | *Local Server* in *Server Manager* window | *IPv4 address assigned by DHCP, IPv6 enabled*
3. Right-clicked *Ethernet* | *Properties*
4. Unselected *Internet Protocol Version 6 (TCP/IPv6)*

5. Double-clicked *Internet Protocol Version 4 (TCP/IPv4)*
6. Clicked *Use the following IP address*
7. Typed 44.38.3.67 in *IP Address* field
8. Inputted 255.255.255.0 in *Subnet mask* field
9. Typed 44.38.3.1 in *Default gateway* field
10. Inputted 44.38.3.31 in *Preferred DNS server* field
11. Clicked **OK** in *Internet Protocol Version 4 (TCP/IPv4) Properties* window | **OK** in *Ethernet Properties* window | *Close* in *Network Connections* window

G03SRV05 Computer Name and AD Domain Configurations

The section detailed how G03SRV05 was enrolled from a workgroup to the group03.c24200.cit.lcl domain. The process allowed the server to participate in domain-based security, Group Policy, and centralized management. Domain enrollment provided the centralized security and management necessity, which ensured consistent user access and organizational control over all network assets. **Refer to Appendix B: System and Network Configuration.**

1. Opened G03SRV05 (see section)
2. Clicked *Server Manager* in *Windows Taskbar* window | *Local Server* in *Server Manager* window | *G03WIN2025TEMP* in *Computer name* window
3. Typed G03SRV05 in *Computer name* field
4. Clicked **OK** in *System Properties* window
5. Typed G03SRV05 in *Computer name* field
6. Inputted group03.c24200.cit.lcl in *Domain* field
7. Clicked **OK** in *Computer Name/Domain Changes* field

8. Inputted Administrator in *User name* field
9. Typed [Group03 Administrator's password] in *Password* field
10. Clicked **OK** in *Windows Security* window | **OK** in *Computer Name/Domain Changes* window | **OK** in *You must restart your computer to apply these changes* window | **Close** in *System Properties* window | **Restart Now** in *Microsoft Windows* window

G03SRV05 Domain Controller Promotion

The section detailed the promotion of G03SRV05 to an AD DC within the group03.c24200.cit.lcl domain. The objective was completed through navigation within the features of Active Directory server manager, and the use of the parent DC administrator credentials during installation, so the prerequisite checks can run successfully. Domain controller abilities were necessary on server G03SRV05, so FSMO roles could later be transferred to the DC. The following steps were documented on G03SRV03.

1. Opened Prism Central (see section)
2. Clicked *Compute* | *VMs* | *G03SRV05* | **Launch Console** | **Send CtrlAltDel** in *G03SRV05 - Profile 1 - Microsoft Edge* window | *Other user*
3. Typed Group03\Administrator in *User name* field
4. Inputted [Group03 Administrator's password] in *Password* field
5. Pressed **Enter**
6. Clicked *Server Manager* in *Windows Taskbar* | *Manage* | *Add Roles and Features* | **Next** in *Before You Begin* window | **Next** in *Installation Type* window | **Next** in *Server Selection* window
7. Selected *Active Directory Domain Services*

8. Clicked **Add Features** in *Add Roles and Features Wizard* window | **Next** in *Server Roles* window | **Next** in *Features* | **Next** in *AD DS* | **Install** | **Close** in *Results* window |

Notifications (icon)

9. Selected *Promote this server to a domain controller* | **Next** in *Deployment Configuration* window
10. Typed [new secure password] in *Password* window
11. Inputted [new secure password] in *Confirm password* window
12. Clicked *Site name:* dropdown | *Default-First-Site-Name* | **Next** in *Domain Controller Options* | **Next** in *DNS Options* window | **Next** in *Additional Options* window | **Next** in *Paths* window
13. Selected **Next** in *Review Options* window | **Install** | **Close** in *You're about to Sign out* window

FSMO Roles Transfer on G03SRV01VM to G03SRV05

The section detailed the transfer of FSMO roles from G03SRV01VM server to G03SRV05 server. FSMO roles transferred allowed the new virtual DC to take over AD operations, which included RID Master, Domain Naming Master, and Infrastructure Master functions. As critical "single-master" directory tasks were assigned to the new server, FSMO roles transfer ensured network stability and prevented conflicts in object creation and domain naming. **Refer to Appendix B: System and Network Configuration.**

1. Clicked *Server Manager* in *Windows Searchbar* | *Tools* | *Active Directory Users and Computers*
2. Right-clicked *Active Directory Users and Computers*

3. Clicked *Change Domain Controller...* | *G03SRV05.group03.c24200.cit.lcl* | **OK** in *Change Directory Server* window
4. Right-clicked *group03.c24200.cit.lcl*
5. Clicked *Operations Masters...* | **Change...** | **Yes** in *Active Directory Domain Services* | **OK** in *Active Directory Domain Services* window |
6. Selected *Infrastructure* | **Change...** | **Yes** in *Active Directory Domain Services* window | **OK** in *Active Directory Domain Services* window | **Close** in *Operations Masters* window
7. Typed `powershell` in *Windows Searchbar* field
8. Right-clicked *Powershell*
9. Clicked *Run as administrator*
10. Typed `$Server = Get-ADDomainController -Identity "G03SRV05.group03.c24200.cit.lcl";
Move-ADDirectoryServerOperationMasterRole -Identity $Server -OperationMasterRole DomainNamingMaster`
11. Pressed **Enter**
12. Typed `repadmin /replsummary; repadmin /showrepl` in *PS*
C:\Users\Administrator field
13. Pressed **Enter**

Category Creation for Windows 11 VMs on Nutanix PC

The section detailed how Nutanix Categories were created and used to logically divide the Windows 11 VM clients workstations from the Windows Server 2025 VM clients. The setup enabled Workload Isolation, which in turn allowed the organization to enforce different security policies and resource priorities for each operating system tier. As the assets were categorized based on the operating system categories, the administration could ensure workstation traffic remained consistent and the specific desktop updates wouldn't disrupt the server environment's high-availability needs, which also allowed for more detailed reports and audits essential for compliance. **Refer to Appendix B: System and Network Configuration.**

1. Clicked *Administration* dropdown | *Categories* | **New Category**
2. Typed `Windows11` in *Name* field
3. Inputted `WIN11` in *Values* field
4. Clicked **Save** in *General* | *Compute* dropdown | *VMs* | *G03SIWIN11VM* | **Update** | **Next** | **Next**
5. Typed `Windows11:WIN11` in *Categories* field
6. Clicked *Windows11:WIN11* | **Save**
7. Repeated steps 5-7 for all Windows 11 VMs

Category Creation for Windows Server 2025 VMs on Nutanix PC

The section detailed how Nutanix Categories were created and used to logically divide the Windows Server 2025 VM clients workstations from the Windows 11 VM clients. The setup enabled Workload Isolation, which in turn allowed the organization to enforce different security policies and resource priorities for each operating system tier. As the assets were categorized based on the operating system categories, the administration could ensure workstation traffic remained consistent and the specific desktop updates wouldn't disrupt the server environment's high-availability needs, which also allowed for more detailed reports and audits essential for compliance. **Refer to Appendix B: System and Network Configuration.**

1. Clicked *Administration* dropdown | *Categories* | **New Category**
2. Typed `WindowsServer2025` in *Name* field
3. Inputted `SRV` in *Values* field
4. Clicked **Save** in *General* window | *Compute* dropdown | *VMs* | `G03SRV03VM` | **Update** | **Next** | **Next**
5. Typed `WindowsServer2025:SRV` in *Categories* field
6. Clicked `WindowsServer2025:SRV` | **Save**
7. Repeated steps 5-7 for all Windows Server 2025 VMs

Roles Creation on Nutanix PC

The section detailed the creation of custom administrative roles within Nutanix PC to implement the Principle of Least Privilege. The process was essential for the organization to differentiate between "Virtual Admins," who had the authority to provision and manage new resources, and "Virtual Operators," who were restricted to existing infrastructure management permission only. As the granular roles, VirtualCreate and VirtualOperator, were created, the enterprise could safely delegate tasks to specific Active Directory users, which ensured team members had exactly the level of access required for operation and prevented unauthorized or accidental changes to the virtual environment. **Refer to Appendix B: System and Network Configuration.**

1. Clicked *Infrastructure* dropdown | *Admin Center* | *IAM* | *Roles* | **Create Role** | *From existing role*
2. Typed Virtual Machine Operator in *Select roles* field
3. Clicked *Virtual Machine Operator* | **Next** in *Choose Role* window | **Continue** in *System Operations* window | **Skip for now**
4. Typed VirtualOperator in *Role Name* field
5. Clicked **Save & Create Authorization Policy** | **Skip for now** | *Close* in *Create New Authorization Policy* window | **Create Role** | *From existing role*
6. Typed Virtual Machine Admin in *Select roles* field
7. Clicked *Virtual Machine Admin* | **Next** in *Choose Role* window | **Continue** in *System Operations* window | **Skip for now**
8. Typed VirtualCreate in *Role Name* field
9. Clicked **Save & Create Authorization Policy** | **Skip for now**

Projects Creation for Windows 11 on Nutanix PC

The section detailed the creation of a dedicated Project within Nutanix Prism Central to establish a segregated management boundary for Windows 11 resources. The process provided a secure, "sandboxed" space where specific virtual machines and infrastructure resources are isolated from the rest of the enterprise environment. As the `Windows11VMConfigure` project was defined and mapped it to specific clusters and VLANs, the organization could enforce resource quotas and dedicated network access, which ensured workstation management tasks do not compete for resources or conflict with critical server infrastructure. The default setting for Directory or Provider is `G03SRV01LDAP`. **Refer to Appendix B: System and Network Configuration.**

1. Clicked *Infrastructure* dropdown | *Admin Center* | *Projects* | + **Create Project**
2. Typed `Windows11VMConfigure` in *Project Name* field
3. Clicked **Create** in *Create Project* window | *Projects* | *exclamation mark icon* in *Windows11VMConfigure* window | + *Add Infrastructure* | **Add Infrastructure** | *NTNX_LOCAL_AZ* | **Configure Resources**
4. Selected `G03NUTXSRV01` in *Select clusters to be added to this project* window | *G03NUTXSRV01* | *LAN1903*
5. Clicked **Confirm and Select Default** in *Select Clusters and VLANs* window | **Confirm** in *Confirm and Select Default* window | **Save** in *Windows11VMConfigure* window

Projects Creation for Windows Server Configuration on Nutanix PC

The section detailed the creation of a dedicated Project within Nutanix Prism Central to establish a segregated management boundary for Windows Server 2025 resources. The process provided a secure, "sandboxed" space where specific virtual machines and infrastructure resources are isolated from the rest of the enterprise environment. As the WindowsServer2025Configure project was defined and mapped it to specific clusters and VLANs, the organization could enforce resource quotas and dedicated network access, which ensured workstation management tasks do not compete for resources or conflict with critical server infrastructure. The default setting for Directory or Provider is G03SRV01LDAP. **Refer to Appendix B: System and Network Configuration.**

1. Clicked *Infrastructure* dropdown | *Admin Center* | *Projects* | + **Create Project**
2. Typed WindowsServer2025Configure in *Project Name* field
3. Clicked **Create** in *Create Project* window | *Projects* | *exclamation mark icon* in *WindowsServer2025Configure* window | + *Add Infrastructure* | **Add Infrastructure** | *NTNX_LOCAL_AZ* | **Configure Resources**
4. Selected *G03NUTXSRV01* in *Select clusters to be added to this project* window | *G03NUTXSRV01* | LAN1903
5. Clicked **Confirm and Select Default** in *Select Clusters and VLANs* window | **Confirm** in *Confirm and Select Default* window | **Save** in *WindowsServer2025Configure* window

Projects Creation for new VM Creation on Nutanix PC

The section detailed how a dedicated, empty project container was created. The process facilitated autonomous resource management and allowed for Self-Service Provisioning within a controlled environment. Self-Service Provisioning allowed departmental teams to autonomously manage resources within a secure, pre-configured environment. As the NewVMCreate project was established, the organization created a separate management space. Here, assigned users, who would be given the "Virtual Admin" role, could independently create and manage private virtual machines, which was done without interference being risked or conflicted with existing production servers. The approach balanced developer agility with infrastructure security with administrative access limitation set to a specific, pre-approved resource pool. **Refer to Appendix**

B: System and Network Configuration.

1. Clicked *Infrastructure* dropdown | *Admin Center* | *Projects* | + **Create Project**
2. Typed NewVMCreate in *Project Name* field
3. Clicked **Create** in *Create Project* window | *Projects* | *exclamation mark icon* in *WindowsServer2025Configure* window | + *Add Infrastructure* | **Add Infrastructure** | *NTNX_LOCAL_AZ* | **Configure Resources**
4. Selected *G03NUTXSRV01* in *Select clusters to be added to this project* window | *G03NUTXSRV01* | *LAN1903*
5. Clicked **Confirm and Select Default** in *Select Clusters and VLANs* window | **Confirm** in *Confirm and Select Default* window | **Save** in *NewVMCreate* window

HR Authorization Policy Creation on Nutanix PC

The section detailed the creation of the HR Authorization Policy, which applied the VirtualOperator role to the Human Resources department's user groups. The policy granted members of the HR_Mod_DL and HR_Read_DL security groups the ability to monitor and manage existing Windows Server 2025 resources within the WindowsServer2025Configure project. As user's scope was restricted via categories and project filters, the organization ensured HR staff could oversee the HR department's infrastructure health without the need of access to sensitive systems in other departments. **Refer to Appendix B: System and Network**

Configuration.

1. Clicked *Infrastructure* dropdown | *Admin Center* | *IAM* | *Authorization Policies* | +
Create Authorization Policy
2. Typed VirtualOperator in *Select Role* field
3. Clicked *VirtualOperator* | **Next** in *Choose Role* window | *Configure access: select entity types & instances* | *Entity Type* dropdown | *AHV VM* | *Filters* | *In Category* | *Edit in Authorization Policy 1* field
4. Inputted Accounting Policy 1 in *Authorization Policy 1* field
5. Typed WindowsServer2025VM: SRV in *Search* field
6. Clicked *WindowsServer2025VM: SRV* | *Filter* dropdown | *In Project*
7. Typed WindowsServer2025VMConfigure in *Search* field
8. Clicked *WindowsServer2025VMConfigure* | **Next** in *Define Scope* window | *Local User* | *G03SRV01LDAP (AD)*
9. Typed HR_Mod_DL (User Groups) in *Please select* field
10. Clicked *HR_Mod_DL (User Groups)*

11. Typed `HR_Read_DL` (User Groups) in *Please select* field
12. Clicked *HR_Read_DL (User Groups)* | **Save** in *Identities* window

Executive Authorization Policy Creation on Nutanix PC

The section detailed the implementation of the Executive Authorization Policy, designed to provide the executive IT support team with visibility into the Windows 11 VMs. The policy mapped the VirtualOperator role to the `Exec_Mod_DL` and `Exec_Read_DL` Active Directory groups. As the policy was applied specifically to the “Windows11VM: WIN11” category within the Windows11VMConfigure project, the organization maintained a secure environment where executive support could perform necessary maintenance on high-level workstations without exposure to core server clusters. **Refer to Appendix B: System and Network Configuration.**

1. Clicked *Infrastructure* dropdown | *Admin Center* | *IAM* | *Authorization Policies* | +
Create Authorization Policy
2. Typed `VirtualOperator` in *Select Role* field
3. Clicked *VirtualOperator* | **Next** in *Choose Role* window | *Configure access: select entity types & instances* | *Entity Type* dropdown | *AHV VM* | *Filters* | *In Category* | *Edit* in *Authorization Policy 1* field
4. Inputted `Executive Policy 1` in *Authorization Policy 1* field
5. Typed `Windows11VM: WIN11` in *Search* field
6. Clicked *Windows11VM: WIN11* | *Filter* dropdown | *In Project*
7. Typed `Windows11VMConfigure` in *Search* field
8. Clicked *Windows11VMConfigure* | **Next** in *Define Scope* window | *Local User* | *G03SRV01LDAP (AD)*
9. Typed `Exec_Mod_DL` (User Groups) in *Please select* field

10. Clicked *Exec_Mod_DL (User Groups)*
11. Typed *Exec_Read_DL (User Groups)* in *Please select* field
12. Clicked *Exec_Read_DL (User Groups)* | **Save** in *Identities* window

Accounting Authorization Policy Creation on Nutanix PC

The section detailed the creation of an elevated Authorization Policy for the Accounting department. The policy assigned the VirtualCreate role, which allowed accounting managers to not only monitor existing systems but also independently provision and manage new virtual machines within the ThirdProject. As the permissions were delegated, the organization increased operational efficiency and agility for the finance team and maintained strict project-level boundaries to protect the rest of the enterprise environment. **Refer to Appendix B: System and Network Configuration.**

1. Clicked *Infrastructure* dropdown | *Admin Center* | *IAM* | *Authorization Policies* | +
Create Authorization Policy
2. Typed *VirtualCreate* in *Select Role* field
3. Clicked *VirtualCreate* | **Next** in *Choose Role* window | *Configure access: select entity types & instances* | *Entity Type* dropdown | *AHV VM* | *Filters* | *In Project* | *Edit in Authorization Policy 1* field
4. Inputted *Accounting Policy 1* in *Authorization Policy 1* field
5. Typed *ThirdProject* in *Search* field
6. Clicked *ThirdProject* | *Filter* dropdown | *On Cluster*
7. Typed *AllCluster* in *Search* field
8. Clicked *AllCluster* | **Next** in *Define Scope* window | *Local User* | *G03SRV01LDAP (AD)*
9. Typed *ACCG_Mod_DL (User Groups)* in *Please select* field

10. Clicked *ACCG_Mod_DL (User Groups)*
11. Typed *ACCG_Read_DL (User Groups)* in *Please select* field
12. Clicked *ACCG_Read_DL (User Groups)* | **Save** in *Identities* window

Administrator Authorization Policy Creation on Nutanix PC

The section detailed the creation of the Administrator Authorization Policy, which provides broad operational visibility to the core Administrators group. The policy ensured the primary administrative team has the VirtualOperator role assigned across the entire physical host (G03NUTXSRV02). The configuration allowed for comprehensive monitoring and troubleshooting processes of all virtual machines in the cluster, which allowed the enterprise's most senior IT staff to maintain system uptime and respond to infrastructure alerts effectively.

1. Clicked *Infrastructure* dropdown | *Admin Center* | *IAM* | *Authorization Policies* | +
Create Authorization Policy
2. Typed *VirtualOperator* in *Select Role* field
3. Clicked *VirtualOperator* | **Next** in *Choose Role* window | *Configure access: select entity types & instances* | *Entity Type* dropdown | *AHV VM* | *Filters* | *On Cluster* | *Edit* in *Authorization Policy 1* field
4. Inputted *Administrator Policy 1* in *Authorization Policy 1* field
5. Typed *G03NUTXSRV02* in *Search* field
6. Clicked *G03NUTXSRV02* | **Next** in *Define Scope* window | *Local User* | *G03SRV01LDAP (AD)*
7. Typed *Administrators (User Groups)* in *Please select* field
8. Clicked *Administrators (User Groups)* | **Save** in *Identities* window

G03SRV06 Deployment on Nutanix PC

The section detailed the deployment of G03SRV06 from the G03WIN2025TEMP template. Deployment from a template ensured a standardized configuration across VMs in the environment and reduced setup time. G03SRV06 Deployment on Nutanix PC provided a centralized management platform and access to the future public, private, and specific web pages. As a future web server, G03SRV06 was essential for business continuity, which provided the high-availability authentication, redundant directory services required to prevent enterprise-wide downtime, and simple user interface (UI) interactions for employees within the infrastructure. **Refer to Appendix B: System and Network Configuration.**

1. Clicked **Create VM From Template** | *Please Select* dropdown | *G03WIN2025TEMP* | **Begin** in *Create VM From Template* window |
2. Typed G03SRV06 in *Name*
3. Clicked **Next** in *Deploy VM from Template* window | **Deploy** in *Deploy VM from template* window | G03SRV06 | *More* dropdown | *Power On* | *Launch Console* | **Send CtrlAltDel** in *G03SRV06* window
4. Inputted [administrator password] in *Password* field
5. Clicked *Local Server* in *Server Manager* | *IPv4 address assigned by DHCP* in *Ethernet* window | *Ethernet* in *Network Connections* window | **Properties** in *Ethernet Status* window
6. Unselected *Internet Protocol Version 6 (TCP/IPv6)* in *Ethernet Properties* window
7. Clicked *Internet Protocol Version 4 (TCP/IPv4)* | **Properties** | *Use the following IP address:* in *Internet Protocol Version 4 (TCP/IPv4) Properties* window
8. Typed 44.38.3.55 in *IP address*

9. Inputted 255.255.255.0 in *Subnet mask*
10. Typed 44.38.3.1 in *Default gateway* field
11. Inputted 44.38.3.31 in *Preferred DNS server* field
12. Clicked **OK** in *Internet Protocol Version 4(TCP/IPv4) Properties* window

G03SRV06 Computer Name and AD Domain Configurations

The section detailed how G03SRV06 was enrolled from a workgroup to the group03.c24200.cit.lcl domain. The process allowed the server to participate in domain-based security, Group Policy, and centralized management. Domain enrollment provided the centralized security and management necessity, which ensured consistent user access and organizational control over all network assets. **Refer to Appendix B: System and Network Configuration.**

1. Clicked *Local Server* in *Server Manager* window | *WORKGROUP* in *Workgroup* window | **Change...** in *System Properties* window
2. Typed G03SRV06 in *Computer name* field
3. Selected *Domain* in *Member of* window
4. Typed group03.c242000.cit.lcl in *Domain* field
5. Clicked **OK** in *Computer Name/Domain Changes* window
6. Inputted Administrator in *User name* field
7. Typed [Group03 Administrator's password] in *Password* field
8. Clicked **OK** in *Windows Security* window
9. Right-clicked **Start**
10. Clicked *Shut down or sign out* | *Restart* | **Continue**

IIS Installation on G03SRV06

The section detailed the activation of the Web Server IIS role on G03SRV06. The process transformed the server into a central host for the organization's web applications and internal documentation. As the Server Manager's "Add Roles and Features" wizard was utilized, the necessary binaries and management tools were deployed to support Hypertext Transfer Protocol (HTTP) traffic, which provided the infrastructure required to host multiple, distinct websites on a single server instance. **Refer to Appendix B: System and Network Configuration.**

1. Clicked *Server Manager* in *Windows Taskbar*
2. Selected *Manage | Add Roles and Features | Next* in *Before You Begin* window | **Next** in *Installation Type* | **Next** in *Server Selection*
3. Selected *Web Server (IIS)* in *Server Roles*
4. Clicked **Add Features** in *Add Roles and Features Wizard* | **Next** in *Server Roles* window | **Next** in *Features* window
5. Selected **Next** in *Web Server Role (IIS)* window | **Next** in *Role Services* window | **Install** in *Confirmation* window | **Close** in *Results* window

Public Webpage Creation on G03SRV06

The section detailed the creation of the physical file structure and HyperText Markup Language (HTML) content for the public web environments. As the directory was organized under C:\inetpub\wwwroot, the organization established a clear general-interest content (Public). The approach to content management ensured web files were staged correctly before the files were mapped to virtual sites within the IIS manager. **Refer to Appendix B: System and Network Configuration.**

1. Clicked *File Explorer* in *Windows Taskbar* | *This PC* | *Local Disk (C:)* | *inetpub* | *wwwroot* | *New Folder* in *Home Toolbar*
2. Typed `Public` in *New Folder Name* field
3. Clicked *Notepad* in *Windows Taskbar*
4. Typed `<html><h1>Public Page</h1><p>Welcome! Anyone can access this page.</p></html>` in *Notepad*
5. Pressed **Ctrl+S**
6. Typed `pubPage.html` in *File name* field
7. Clicked *This PC* | *Local Disk (C:)* | *inetpub* | *wwwroot* | *Public* | **Save**

Private Webpage Creation on G03SRV06

The section detailed the creation of the physical file structure and HTML content for the private web environments. As the directory was organized under `C:\inetpub\wwwroot`, the organization established a clear internal department data (Private). The approach to content management ensured web files were staged correctly before the files were mapped to virtual sites within the IIS manager. **Refer to Appendix B: System and Network Configuration.**

1. Opened *File Explorer*
2. Clicked *This PC* | *Local Disk (C:)* | *inetpub* | *wwwroot* | *New Folder* in *Home Toolbar* window
3. Typed `Private` for *New Folder Name*
4. Clicked *Notepad* in *Windows Taskbar*
5. Typed `<html><h1>Private Page</h1><p>Welcome! This is a private page.</p></html>` in *Notepad* window
6. Pressed **Ctrl+S**

7. Typed `privPage.html` in *File name* field
8. Clicked *This PC | Local Disk (C:) | inetpub | wwwroot | Private | Save*

Site Specific Webpage Creation on G03SRV06

The section detailed the creation of the physical file structure and HTML content for the site specific web environments. As the directory was organized under `C:\inetpub\wwwroot`, the organization established clear, restricted, and site-specific resources (SiteSpec). The approach to content management ensured web files were staged correctly before the files were mapped to virtual sites within the IIS manager. **Refer to Appendix B: System and Network**

Configuration.

1. Opened *File Explorer*
2. Clicked *This PC | Local Disk (C:) | inetpub | wwwroot | New Folder* in *Home Toolbar* window
3. Typed `SiteSpec` for *New Folder Name*
4. Clicked *Notepad* in *Windows Taskbar*
5. Typed `<html><h1>Site Specific Page</h1><p>Welcome! This is a Site Specific page.</p></html>` in *Notepad* window
6. Pressed **Ctrl+S**
7. Typed `sitePage.html` in *File name* field
8. Clicked *This PC | Local Disk (C:) | inetpub | wwwroot | SiteSpec | Save*

IIS Websites and Access Control Configurations for Public and Private Websites on G03SRV06

The section detailed the configuration of virtual websites within IIS to host multiple services on a single IP address (44.38.3.55). For the business case, the process demonstrated efficient resource utilization. As the PublicSite was assigned to the standard HTTP port (80) and the PrivateSite to port 8080, the organization could serve different audiences from a single server without the need of additional network interfaces or IP addresses. **Refer to Appendix B: System and Network Configuration.**

1. Clicked *Server Manager* in *Windows Taskbar* | *Tools* | *Internet Information Services (IIS) Manager* | *G03SRV06* in *Connections* window
2. Right-clicked *Sites*
3. Clicked *Add Website...*
4. Typed *PublicSite* in *Site name* field
5. Inputted *C:\inetpub\wwwroot\Public* in *Physical path* field
6. Clicked *All Unassigned* in *IP address* window | *44.38.3.55* in *IP address* window | **OK** in *Add Website* window
7. Right-clicked *Sites*
8. Clicked *Add Website...*
9. Typed *PrivateSite* in *Site name* field
10. Inputted *C:\inetpub\wwwroot\Private* in *Physical path* field
11. Clicked *All Unassigned* in *IP address* window | *44.38.3.55* in *IP address* window
12. Typed *8080* in *Port* field
13. Clicked **OK** in *Add Website* window

IIS Websites and Access Control Configurations for Specific Website on G03SRV06

The section detailed the configuration of virtual websites within IIS to host multiple services on a single IP address (44.38.3.55). For the business case, the process demonstrated efficient resource utilization. As the SpecSite was assigned to port 8081, the organization could serve different audiences from a single server without the need of additional network interfaces or IP addresses. **Refer to Appendix B: System and Network Configuration.**

1. Clicked *Server Manager* in *Windows Taskbar* | *Tools* | *Internet Information Services (IIS) Manager* | *G03SRV06* in *Connections* window
2. Right-clicked *Sites*
3. Clicked *Add Website...*
4. Typed SpecSite in *Site name* field
5. Inputted C:\inetpub\wwwroot\SiteSpec in *Physical path* field
6. Clicked *All Unassigned* in *IP address* window | *44.38.3.55* in *IP address* window
7. Typed 8081 in *Port* field
8. Clicked **OK** in *Add Website* window

Authentication Configuration for Private and Specific Site on G03SRV06

The section detailed how the hardening process of the Private and Site-Specific websites was moved beyond anonymous access. The process was critical to protect sensitive organizational data. As Anonymous Authentication was disabled and Windows Authentication was enabled, the server forced users to identify themselves with the use of AD credentials. Furthermore, the application of Authorization Rules ensured only members of specific groups, such as Domain Users or Site Users, could view the content, which effectively enforced the principle of least privilege. **Refer to Appendix B: System and Network Configuration.**

1. Clicked *Server Manager* in *Window Taskbar* | *Tools* | *Internet Information Services (IIS) Manager* | *G03SRV06* in *Connections* window | *Sites* | *PrivateSite* | *Authentication* in *Home* window | *Anonymous Authentication* in *Authentication* window
2. Selected *Disable* in *Actions* window | *Windows Authentication* in *Authentication* window
3. Clicked *Enable* in *Actions* window | *PrivateSite* in *Connections* window | *Authorization Rules* in *Home* window | *Add Allow Rule...* in *Actions* window | *Specified roles or user groups* in *Allow access to this Web content to* window
4. Typed `Domain Users` in *Specified roles or user groups* field
5. Clicked **OK** in *Add Allow Authorization Rule* window | *SpecSite* in *Connections* window | *Authentication* in *Home* | *Anonymous Authentication* in *Authentication* window | *Disable* in *Actions* window | *Windows Authentication* in *Authentication* window
6. Selected *Enable* in *Actions* window | *SpecSite* in *Connections* | *Authorization Rules* in *Home* | *Add Allow Rule...* in *Actions* | *Specified roles or user groups* in *Allow access to this Web content to* window
7. Typed `Site Users` in *Specified roles or user groups* field
8. Clicked **OK** in *Add Allow Authorization Rule* window

IP Restrictions Configuration for SpecSite on G03SRV06

The section detailed the implementation of IP Address and Domain Restrictions for the SpecSite. The configuration provided an additional layer of "Defense-in-Depth" with restricted access configuration at the network level. For the business case, the process ensured: even with valid credentials, the site was only accessible to devices, which resided within the trusted 44.38.3.0/24 subnet. As the default action was set to "Deny" for unspecified clients, the organization significantly reduced the attack surface for the specific web resource. **Refer to Appendix B: System and Network Configuration.**

1. Clicked *Server Manager* in *Windows Taskbar* | *Tools* | *Internet Information Services (IIS) Manager* | *G03SRV06* in *Connections* | *Sites* | *SpecSite* | *IP Address and Domain Restrictions* in *Home* window | *Edit Feature Settings* in *Actions* window
2. Selected *Deny* in *Access for unspecified clients* window
3. Clicked **OK** in *Edit IP and Domain Restrictions Settings* window | *Add Allow Entry* in *Actions* window | *IP address range* in *Allow access for the following IP address or domain name* window
4. Typed 44.38.3.0 in *IP address range* field
5. Inputted 255.255.255.0 in *Mask or Prefix* field
6. Clicked **OK** in *Add Allow Restriction Rule* window

Firewall Configuration on G03SRV06

The section detailed the programmatic configuration of the Windows Defender Firewall to allow traffic on non-standard ports. Since Windows Server defaulted to block inbound traffic on custom ports like 8080 and 8081, the PowerShell commands were essential to ensure the new web services were reachable by clients across the network. As the necessary protocols were allowed and the default block on other ports was maintained, the organization managed a strong security posture where necessary business communications were enabled. **Refer to Appendix B: System and Network Configuration.**

1. Typed `Powershell` in *Search Bar*
2. Right-clicked *Powershell*
3. Clicked *Run as administrator*
4. Typed `New-NetFirewallRule -DisplayName "Allow HTTP 8080"`
`-Direction Inbound -Protocol TCP -LocalPort 8080 -Action`
`Allow` in *C:\Users\System32* field
5. Pressed **Enter**
6. Typed `New-NetFirewallRule -DisplayName "Allow HTTP 8081"`
`-Direction Inbound -Protocol TCP -LocalPort 8081 -Action`
`Allow` in *C:\Users\System32* field
7. Pressed **Enter**

RESULTS

The current WoodRock enterprise environment was expanded into a multi-site environment with multiple subnets, virtualized access control, function backups with storage pools, and multi-scope web hosting. The additions to the environment aligned with the business scenario as reliability, access management, and data protection were improved.

The network architecture was improved by the creation of the LAN1903 subnet, which allowed for a second site within the organization. As a second site was implemented, Active Directory Sites and Services associated the 44.38.103.0/24 subnet with GROUP03SITE, which showed cross site authentication. Clients located within the second site now authenticated against local domain controllers, which improved reliability and decreased traffic between sites.

Domain redundancy was achieved by the deployment of G03SRV04 and G03SRV05 and the two servers' promotion to domain controllers. FSMO roles were transferred to G03SRV05, which allowed for multi-site redundancy. The distribution of FSMO roles to different domain controllers ensured operations were still functional in case a domain controller failed. Replication between domain controllers was configured correctly, which ensured data was consistent across both sites.

Servers and workstations were enhanced through access control in Prism Central, where roles were created to organize users and groups with the use of LDAP authentication. Role based access control was implemented through the use of custom roles such as VirtualOperator and VirtualCreate. Other roles included Windows11VMConfigure, WindowsServer2025Configure, and NewVMCreate were implemented to allow for more fine grained access control between users and groups. New authorization policies were assigned to departments within WoodRock, which included Human Resources, Accounting, Executive, and Administrators, based on scope

and group membership; hence, the authorization policies ensured controlled access to the virtual environment.

Centralized storage for backups was created through the deployment of G03TRUENAS, which provided dedicated storage for backups. The storage pool G03NAS was created with two 250GB virtual disks. Datasets GeneralSet03 and BackupSet03 were created within G03NAS and configured to separate general data from backup data. Server Message Block was configured with access control lists, which allowed only administrative accounts to access BackupSet03.

Backups were configured and scheduled on G03SRV01VM with the use of Windows Server Backup. Along with the two scripts written to rename and delete backups, backups were configured to store only the system state and user data to the TrueNAS share. Task Scheduler was implemented for script automation run within the G03SRV01VM system.

A IIS web server was deployed within the G03SRV06 virtual machine to allow access to multiple websites with different scopes. Domain and site specific webpages were created and enforced, which operated on separate ports and allowed for correct domain and site authentication for specific users.

The final WoodRock enterprise environment operated without any permission conflicts or backup failures. The multi-site environment showed proper site replication and authentication, along with proper redundancy configurations with the distribution of various FSMO roles. Properly configured role based access control within a virtual environment allowed for appropriate access for every user within the domain. Centralized storage and correctly configured backups allowed for secure data management and integrity. The completed Woodrock infrastructure showed proper scalability and security as defined in the business scenario.

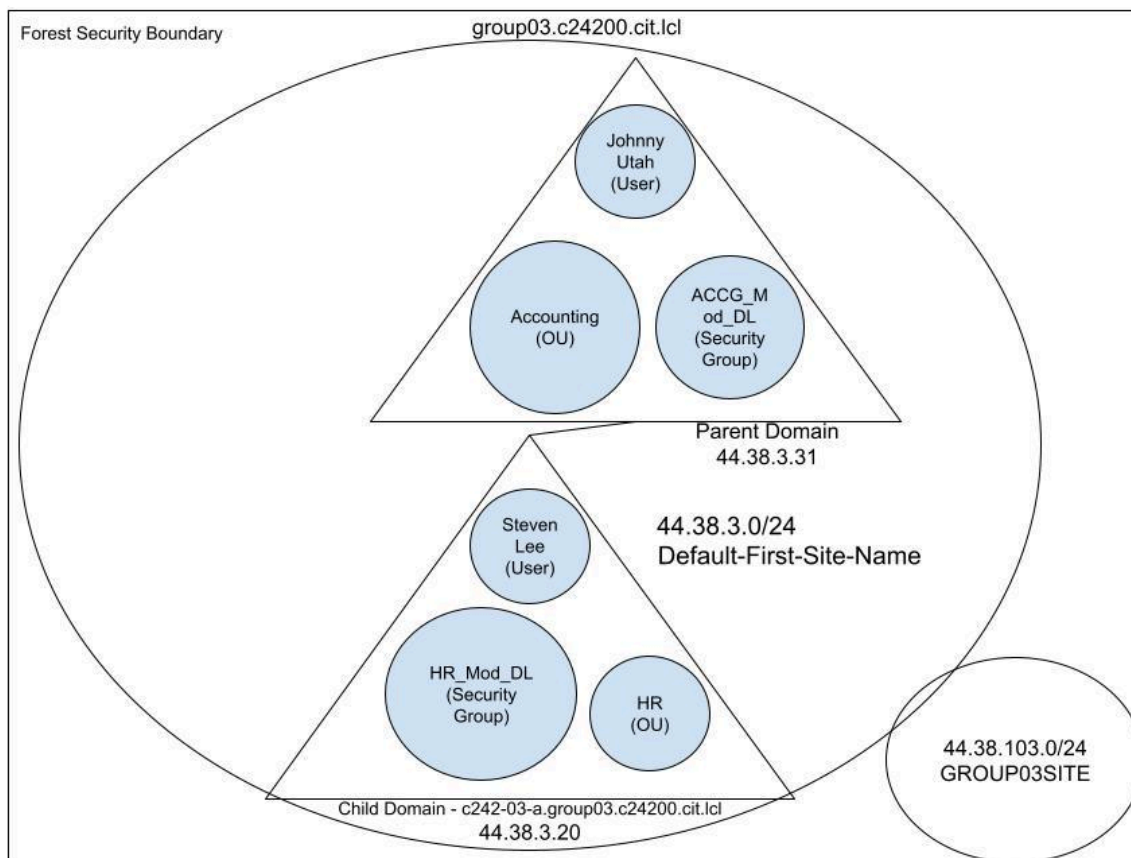


Figure 3.) Logical model of WoodRock's forest

WoodRock had decided to keep the domain structure relatively the same since the incorporation of the most recent advancements. Site GROUP03SITE was added to replicate between, which gave WoodRock more leniency with the physical location of operations. The site was tied to a newly instated subnet, under 44.38.103.0/24, which opened job positions in site-specific VM management. With no surprise, Steven Lee and Johnny Utah paved the way for young WoodRock employees, which left no reason to tamper with any department or OU. For now, the parent and child domain trust remained intact, with no additional domain had been added.

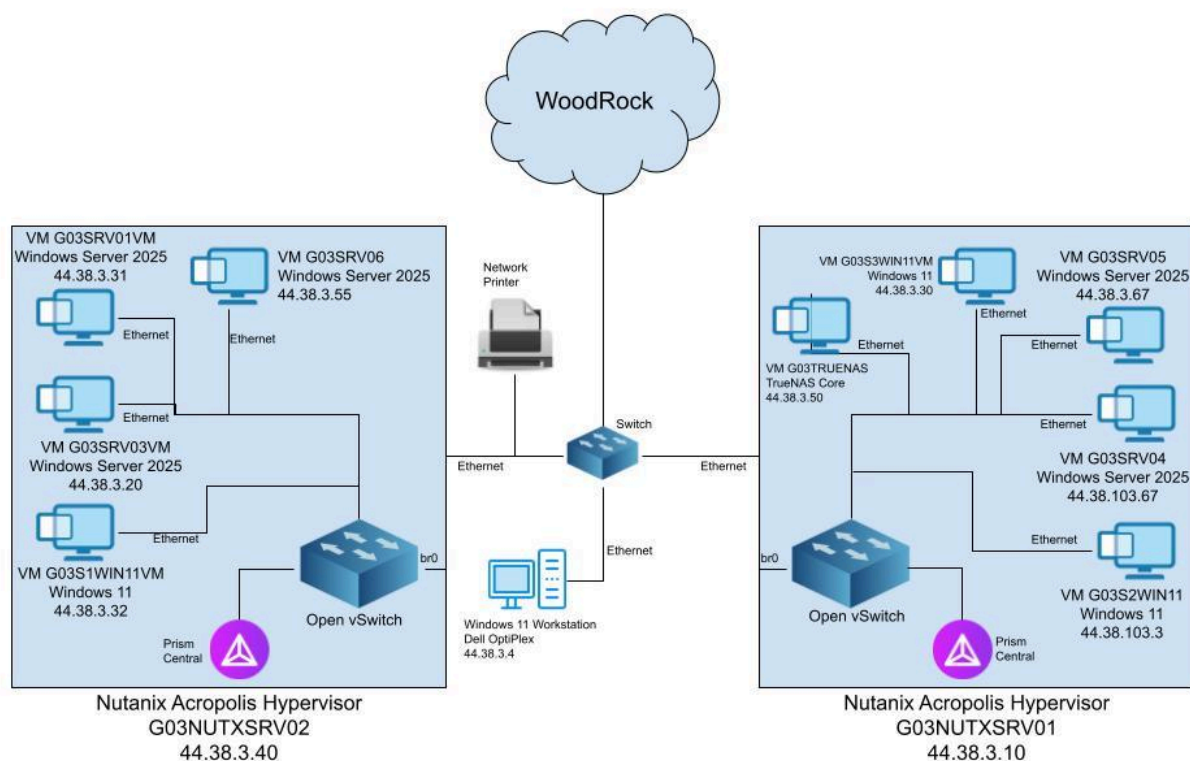


Figure 4.) Physical model of WoodRock topology

WoodRock did not hold back as the company's outreach of virtual machines advanced, as five more were configured across both clusters G03NUTXSRV01 and G03NUTXSRV02. The virtual machines G03SRV05, G03SRV04, G03S2WIN11, and G03TRUENAS had been launched on G03NUTXSRV01, while G03SRV06 was launched on G03NUTXSRV01. WoodRock had decided to transfer G03S3WIN11VM from G03NUTXSRV02 to G03NUTXSRV01 via live migration. As mentioned, the company incorporated a new subnet to inherit the G03SRV04 and G03S2WIN11 virtual machines, which gave WoodRock an outlet to increase security and lower network traffic. WoodRock kept the overall ethernet topology intact, all supported by the WoodRock/CIT network connection.

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APPENDIX A: PROBLEM SOLVING

The problem solving section provided a comprehensive overview of problems faced during the configuration of all systems and virtual networks in the enterprise windows server administration project. The first problem faced during the configuration of the enterprise environment was the TrueNAS backup share was not accessible from the windows server backup, as a destination could not be accessed. The second problem faced in the project happened when the IIS website incorrectly restricted users; consequently, websites failed to load. The third problem faced in the project was the failure of FSMO role transfer for the configuration of the new domain controller, where the failure prevented proper distribution of Active Directory roles across the domain.

Problem 1: TrueNAS Backup Inaccessibility in Windows Server Backup

Problem Description: While the configuration of Windows Server Backup G03SRV01VM took place, the network share \\44.38.3.50\BackupSet03 could not be accessed when a backup destination was selected. The system could not recognize the UNC path as a valid path and failed to connect to the TrueNAS dataset, which inhibited the configuration of backups.

Possible Solutions: Many possible causes of the problem were considered. Incorrect DNS configuration was considered as the TrueNAS system also needed to point to the DNS. The DNS solution would work because the system needed to connect to the domain through both DNS and the UNC path. The SMB and ACL permissions reconfiguration, which could allow necessary users with access to the dataset, was a possible solution. A plausible explanation was because every system was incorrectly blocked from access to the dataset, which would make the UNC path unusable. The final possible solution was

confirmed as the TrueNAS IP was pinged; hence, the trouble shooting process confirmed network connection issues, which would make the dataset unreachable from any system.

Solutions Attempted: Proper Network connectivity was first verified as the TrueNAS IP was pinged through command prompt; as a result, the troubleshooting process had confirmed access to the IP was not the problem.

Final Solution: The issue was finally resolved as the DNS configuration was properly corrected on TrueNAS and G03SRV01VM to ensure proper name resolution. The DNS solution worked thanks to the proper configuration, which allowed the two systems to locate each other and authenticate through SMB. The proper configuration proved the other potential causes failed as the functions were already configured correctly.

Problem 2: IIS Websites Incorrectly Restricted

Problem Description: While the IIS web server on G03SRV06 was tested, the two private hosted websites failed to load due to the improper permissions. Improper roles were defined, which made the websites inaccessible to all users, no matter the site or domain.

Possible Solutions: Firewall configuration was evaluated as a possible cause because blocked ports would make the intended server unreachable; however, updated firewall rules would work as new ports could be reached without restriction. Authentication settings were a possible cause as incorrect configuration could allow unauthorized users to deny allowed users, which correctly configured authentication settings through IIS Authentication Rules to certain domain and site users could solve the issue. Proper subnet access settings configuration in IIS could solve the issue because users on the intended subnet could not access the website as the subnet was not defined.

Solutions Attempted: Firewall rules were examined and updated to allow traffic on ports 80, 8080, and 8081, which confirmed external requests could reach the intended server.

Authentication settings were configured according to site specifications and requirements, which ensured proper users and sites could access the intended website.

Final Solution: The issue was solved after the firewall rules were updated. As the commands, 'New-NetFirewallRule -DisplayName "Allow HTTP 8080" -Direction Inbound -Protocol TCP -LocalPort 8080 -Action Allow' and 'New-NetFirewallRule -DisplayName "Allow HTTP 8081" -Direction Inbound -Protocol TCP -LocalPort 8081 -Action Allow', were typed in C:\Users\System32 field, traffic was able to reach the designated ports. The commands worked because the previous requests were not allowed through network controls, and due to the commands, the ports were opened to incoming traffic.

Problem 3: FSMO Role Transfer Failure

Problem Description: The transfer of FSMO roles from G03SRV01VM to G03SRV05 initially failed. The failure of the transfer of roles prevented proper distribution of Active Directory roles across an enterprise environment, which impacted redundancy and fault tolerance. Without successful role transfer and configuration, crucial roles all remained on a single domain controller, which inhibited redundancy.

Possible Solutions: One solution to the problem was insufficient permissions were given to the administrator of G03SRV05, as a user, must be in certain groups to be able to have certain FSMO roles; hence, the user was added to the proper groups such as Domain Naming Master, role, which led to successful transfer. Incorrect target identification of the transfer was considered as one of the factors, so as the target was corrected, roles could successfully be transferred. The final solution considered was to ensure G03SRV05

was recognized as a domain controller and had proper synchronization within the domain, which would be a viable solution as FSMO roles must be transferred to a domain controller.

Solutions Attempted: One solution attempted was to send a ping request to G03SRV05 to ensure the domain controller could be reached from G03SRV01VM. Although the ping was successful, the problem was not resolved. G03SRV05 was also recognized as a domain controller, which proved the cause of the problem was not due to improper promotion.

Final Solution: The solution which allowed for the transfer of FSMO roles to G03SRV05 where the administrator account was added to the Domain Naming Master Group. As the administrative account was added to the proper group, the domain controller was given permission to operate as the Domain Naming Master. By the successful transfer of FSMO roles, the problem was confirmed to be insufficient permissions and redundancy was achieved.

APPENDIX B: SYSTEM AND NETWORK CONFIGURATION

The following section provided an overview of the new subnet and virtual machines created during the enterprise windows server administration project. The following tables outlined the software specification of each system and recorded how each system was organized for management. All newly created virtual machines, subnets, storage, access control and website configurations are addressed in the tables.

Table 1: VM Configuration

VM configurations and respective roles

VM Name	IP Address	Role
G03SRV01VM	44.38.3.31	Primary Domain Controller, DNS Server, Backup Source
G03SRV03VM	44.38.3.20	Child Domain Controller, LDAP Authentication
G03TRUENAS	44.38.3.50	Centralized Storage Server, SMB Share Host
G03SRV04	44.38.103.2	Secondary Domain Controller for Site 2
G03SRV05	44.38.3.67	Domain Controller, FSMO Role Holder
G03S2WIN11	44.38.103.3	Client Workstation for Site 2
G03SRV06	44.38.3.55	IIS Web Server

Table 2: Subnet Configuration*Subnet configuration of second site*

Subnet Name	Network Address	VLAN ID	Gateway	DNS Server	Purpose
LAN1903	44.38.103.0/24	1903	44.38.103.1	44.38.3.31	Secondary Site, Client and DC Segmentation

Table 3: Storage and Dataset Configuration*Configurations of the datasets within the created storage pool*

Storage System	Storage Pool	Dataset Name	Access Control	Purpose
G03TRUENAS	G03NAS	GeneralSet03	Domain Users	General file storage for users
G03TRUENAS	G03NAS	BackupSet03	Administrators	Backup storage for domain controllers

Table 4: Access Control Configuration*Access control configurations for users and groups through Nutanix IAM*

Role Name	Assigned Group(s)	Project Scope	Permissions
VirtualOperator	HR_Mod_DL, HR_Read_DL	WindowsServer2025 Configure	Monitor and manage existing VMs
VirtualOperator	Exec_Mod_DL, Exec_Read_DL	WindowsServer2025 Configure	Monitor executive workstations
VirtualCreate	ACCG_Mod_DL, ACCG_Read_DL	ThirdProject	Create and manage new VMs
VirtualOperator	Administrators	Cluster-wide	Full monitoring and operational access

Table 5: IIS Website Configuration*Configurations of each website hosted on G03SRV06*

Website Name	Port	Authentication Requirement	Network Restriction	Purpose
Public Site	80	None	None	Public access for general users
Private Site	8080	Domain Authentication	None	Internal access for authenticated users
SpecSite	8081	Domain Authentication	Site 2 Subnet	Restricted access based on subnet

APPENDIX C: BACKUP SCRIPT CONFIGURATION

Scripts written for the enterprise windows server administration project mentioned during the procedure section were included below. The scripts written for the project included a script to rename backups after creation and a script to delete backups older than seven days. The two scripts were written in notepad and saved as windows command script (*.cmd) files.

Table 6: Delete Backup Script*Script written to delete backups after seven days*

```
@echo off
setlocal
net use Z: \\44.38.3.50\BackupSet03 /persistent:no
forfiles /p "Z:\WindowsImageBackup" /m "Backup*" /d -7 /c "cmd /c rd /s /q @path"
net use Z: /delete /yes
```

endlocal

Table 7: Rename Backup Script

Script written to rename backups after creation

```
@echo off

setlocal

set "YY=%date:~-4%"

set "MM=%date:~4,2%"

set "DD=%date:~7, 2%"

set "HH=%time:~0,2%"

set "SS=%time:~3,2%"

if "%HH:~0,1%"==" " set "HH=0%HH:~1,1%"

set "Stamp=%YY%- %MM%- %DD%_ %HH%= %SS%"

ren "\\44.38.3.50\BackupSet03\WindowsImageBackup\G03SRV01VM" "Backup-%Stamp%"

endlocal
```
